Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

Load Dataset

```
df = pd.read_csv('/content/Admission_Predict.csv')  # or your file name
df.head()
```

	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit
0	1	337	118	4	4.5	4.5	9.65	1	0.92
1	2	324	107	4	4.0	4.5	8.87	1	0.76
2	3	316	104	3	3.0	3.5	8.00	1	0.72
3	4	322	110	3	3.5	2.5	8.67	1	0.80
4	5	314	103	2	2.0	3.0	8.21	0	0.65

Fix Column Names

```
df.columns = df.columns.str.strip()
print(df.columns)
```

```
Index(['Serial No.', 'GRE Score', 'TOEFL Score', 'University Rating', 'SOP',
       'LOR', 'CGPA', 'Research', 'Chance of Admit'],
      dtype='object')
```

Drop Unnecessary Column

```
df = df.drop(columns=['Serial No.'])
```

Create Target Variable

```
# Convert to classification (0 or 1)
df['Admitted'] = (df['Chance of Admit'] >= 0.5).astype(int)

# Drop original column
df = df.drop(columns=['Chance of Admit'])
```

Fix Column Names

```
X = df[['GRE Score', 'TOEFL Score', 'University Rating',
        'SOP', 'LOR', 'CGPA', 'Research']]

y = df['Admitted']
```

Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

Decision Tree Model

```
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)
```

```
▾ DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

Predictions

```
y_pred = model.predict(X_test)
```

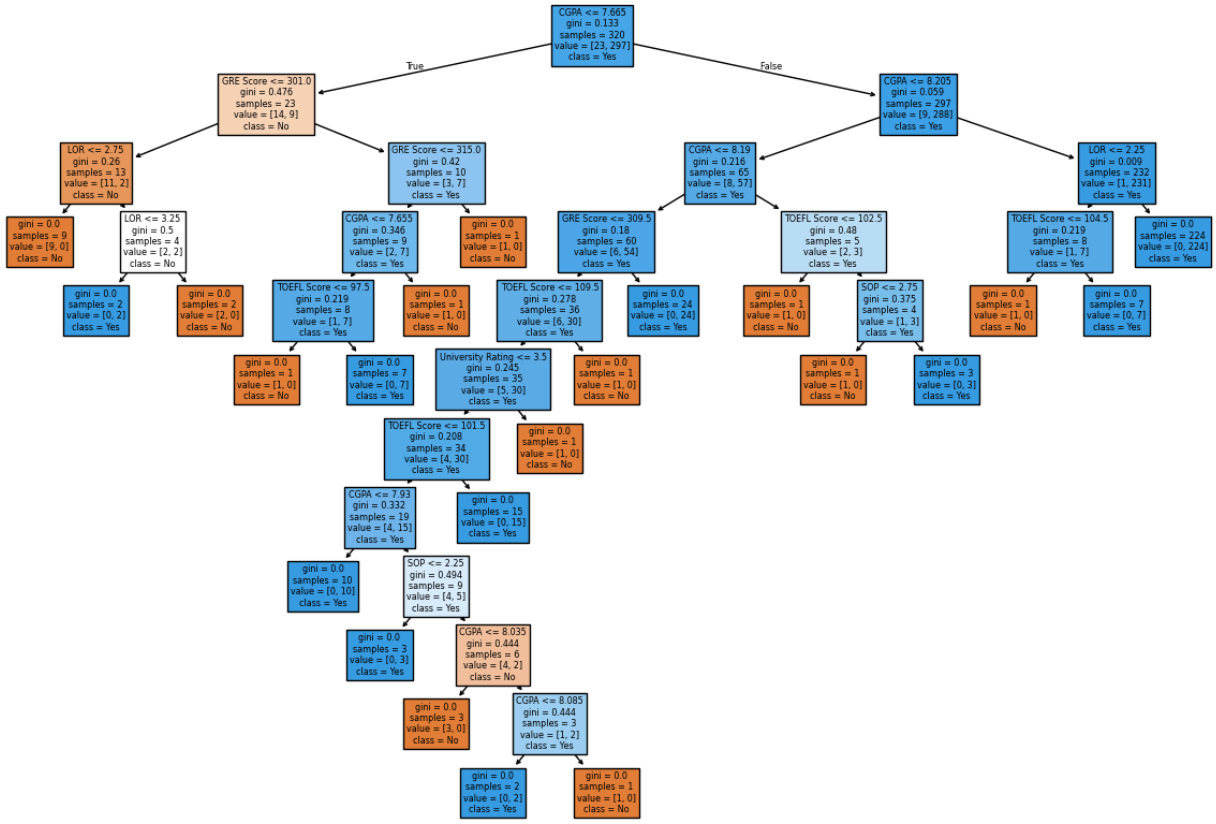
Evaluation

```
print("Accuracy:", accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))

print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.8625

Confusion Matrix:
 [[ 4  6]
 [ 5 65]]

Classification Report:
               precision    recall  f1-score   support

           0       0.44      0.40      0.42        10
           1       0.92      0.93      0.92        70

    accuracy                           0.86        80
   macro avg       0.68      0.66      0.67        80
weighted avg       0.86      0.86      0.86        80
```

Visualize Tree

```
plt.figure(figsize=(15,10))
plot_tree(model, feature_names=X.columns, class_names=['No', 'Yes'], filled=True)
plt.show()
```



Test Custom Student

```
sample = [[320, 110, 4, 4, 4, 9.0, 1]]

pred = model.predict(sample)

if pred[0] == 1:
    print("Admitted ✅")
else:
    print("Not Admitted ❌")
```

```
Admitted ✅
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names,
  warnings.warn(
```